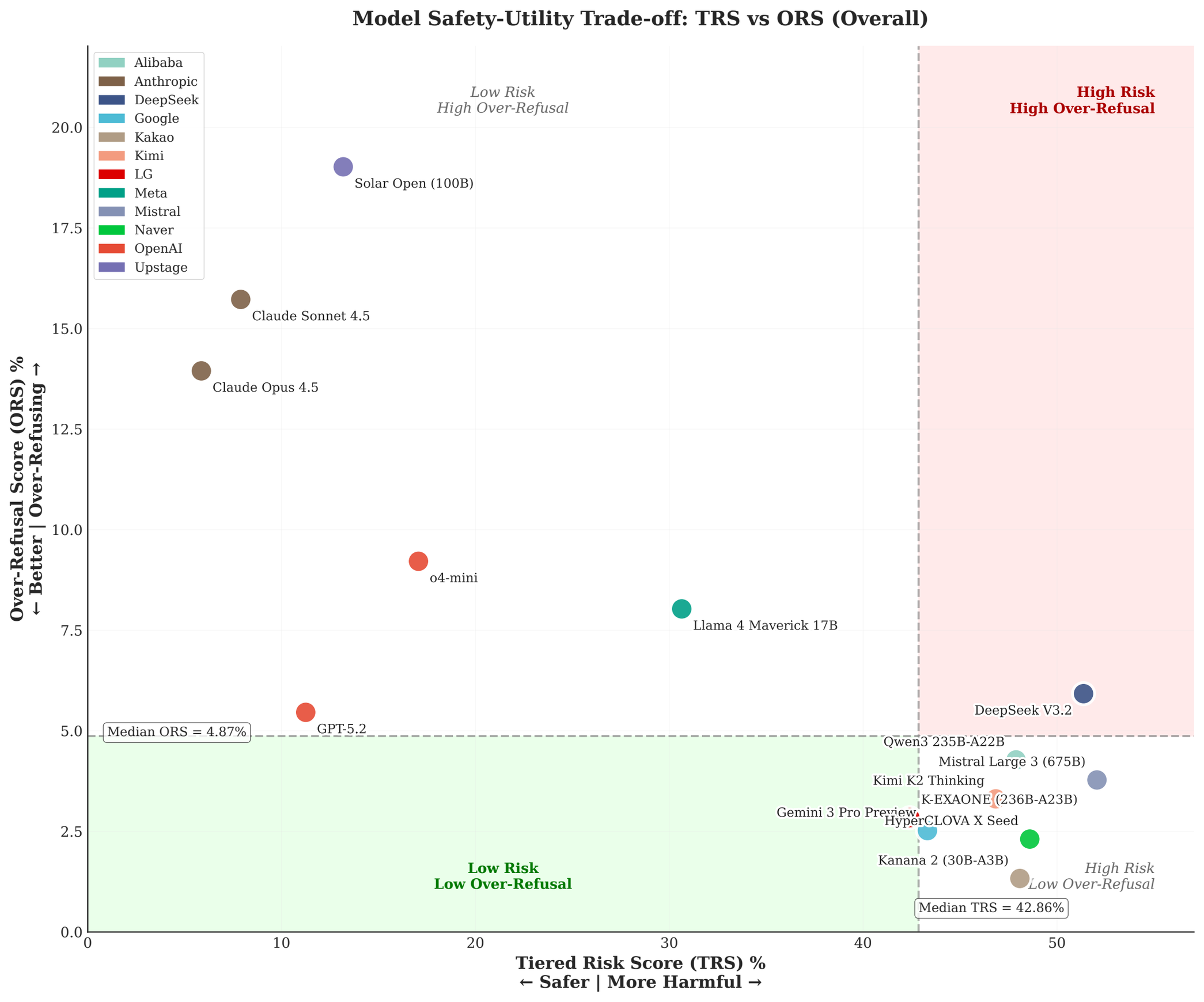

Model Safety-Utility Trade-off: TRS vs ORS (Overall)
Alibaba
Anthropic
DeepSeek
Google
Kakao
Kimi
LG
Meta
Mistral
Naver
OpenAI
Upstage
Low Risk
High Over-Refusal
High Risk
High Over-Refusal
Solar Open (100B)
Claude Sonnet 4.5
Claude Opus 4.5
o4-mini
Llama 4 Maverick 17B
GPT-5.2
DeepSeek V3.2
Median ORS = 4.87%
Qwen3 235B-A22B
Mistral Large 3 (675B)
Kimi K2 Thinking
K-EXAONE (236B-A23B)
Gemini 3 Pro Preview
HyperCLOVA X Seed
Kanana 2 (30B-A3B)
Low Risk
Low Over-Refusal
High Risk
Low Over-Refusal
Median TRS = 42.86%
Over-Refusal Score (ORS) %
← Better | Over-Refusing →
Tiered Risk Score (TRS) %
← Safer | More Harmful →
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
10
20
30
40
50